

ACETONE**ICSC: 0087 (April 2024)**

2-Propanone
Dimethyl ketone
Methyl ketone

CAS #: 67-64-1**UN #: 1090****EC Number: 200-662-2**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Highly flammable. Vapour/air mixtures are explosive. Heating will cause rise in pressure with risk of bursting.	NO open flames, NO sparks and NO smoking. Closed system, ventilation, explosion-proof electrical equipment and lighting. Do NOT use compressed air for filling, discharging, or handling. Use non-sparking handtools.	Use powder, alcohol-resistant foam, water, carbon dioxide. In case of fire: keep drums, etc., cool by spraying with water.

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Sore throat. Cough. Headache. Dizziness. Confusion. Drowsiness. Unconsciousness.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Refer for medical attention.
Skin	Dry skin.	Protective gloves.	Remove contaminated clothes. Rinse skin with plenty of water or shower.
Eyes	Redness. Pain. Blurred vision.	Wear safety spectacles.	Rinse with plenty of water (remove contact lenses if easily possible). Refer for medical attention.
Ingestion	See Inhalation.	Do not eat, drink, or smoke during work.	Rinse mouth. Do NOT induce vomiting. Refer for medical attention .

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Remove all ignition sources. Personal protection: filter respirator for organic gases and vapours of low boiling point adapted to the airborne concentration of the substance. Ventilation. Collect leaking liquid in sealable containers. Absorb remaining liquid in sand or inert absorbent. Then store and dispose of according to local regulations.	According to UN GHS Criteria  DANGER Highly flammable liquid and vapour Causes eye irritation May cause respiratory irritation Transportation UN Classification UN Hazard Class: 3; UN Pack Group: II
STORAGE	
Fireproof. Separated from : see Chemical Dangers. Store in an area without drain or sewer access.	
PACKAGING	



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PHYSICAL & CHEMICAL INFORMATION

Physical State; Appearance

COLOURLESS LIQUID WITH CHARACTERISTIC ODOUR.

Physical dangers

The vapour is heavier than air and may travel along the ground; distant ignition possible.

Chemical dangers

Contact with strong oxidants such as acetic acid, nitric acid and hydrogen peroxide generates explosive peroxides. Reacts with chloroform and bromoform under basic conditions. This generates fire and explosion hazard. Attacks plastics.

Formula: C₃H₆O / CH₃-CO-CH₃

Molecular mass: 58.1

Boiling point: 56°C

Melting point: -95°C

Relative density (water = 1): 0.8

Solubility in water: miscible

Vapour pressure, kPa at 20°C: 24

Relative vapour density (air = 1): 2.0

Relative density of the vapour/air-mixture at 20°C (air = 1): 1.2

Flash point: -18°C c.c.

Auto-ignition temperature: 465°C

Explosive limits, vol% in air: 2.2-13

Octanol/water partition coefficient as log Pow: -0.24

Viscosity: 0.34 mm²/s at 40°C

EXPOSURE & HEALTH EFFECTS

Routes of exposure

The substance can be absorbed into the body by inhalation.

Effects of short-term exposure

The substance is irritating to the eyes and respiratory tract. Exposure at high levels could cause lowering of consciousness.

Inhalation risk

A harmful contamination of the air can be reached rather quickly on evaporation of this substance at 20°C, on spraying or dispersing much faster.

Effects of long-term or repeated exposure

The substance defats the skin, which may cause dryness or cracking.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 250 ppm as TWA; 500 ppm as STEL; BEI issued; A4 (not classifiable as a human carcinogen).

EU-OEL: 1210 mg/m³, 500 ppm as TWA

ENVIRONMENT

NOTES

ADDITIONAL INFORMATION

EC Classification

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